



All Three at Once (x, y, z)

Across, up, and forward — together

x, y, z 세 숫자를 한꺼번에 — 어디에나 놓을 수 있어요

BIG PICTURE

One block can sit anywhere



Give all three numbers together, then **fill** builds a whole box.

세 숫자를 같이 주면 공간 어디든 — fill로 상자도 통째로

HOW TO COUNT

Count the axis — all three together

1 Stand on the start block — that spot is $(0, 0, 0)$.

2 **x** → walk **right**, +1 each block.

3 **y** → jump **up**, +1 each block.

4 **z** → walk **forward**, +1 each block.

Now you change **all three** at once. Order is always (x, y, z) .

이제 셋을 동시에 바꿔요 — 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place GOLD BLOCK at (2, 5, 3)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Set all three numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — the block lands at that exact spot

코드 빌더(C) → (x, y, z) 세 숫자 → Run ▶ → 정확히 그 자리에

CONCEPT

Place one block anywhere

```
place GOLD_BLOCK at (2, 5, 3)
```

$x = 2$ across, $y = 5$ up, $z = 3$ forward — all chosen **together**. With nothing under it, the block floats in the air.

세 숫자를 동시에 정하면 공중 어디든 놓여요

CONCEPT

What is `fill`?

Give `fill` **two corner coordinates** — each a full `(x, y, z)`. It fills everything between them. No loops.

```
fill stone from (0, 0, 0) to (4, 3, 4)
```

두 모서리 좌표(각각 x, y, z)만 주면 사이를 다 채워요

CONCEPT

Solid → hollow



fill stone

solid box
(all stone)



fill air

empties the inside
(now a room)

돌로 채우고, 안쪽은 공기(air)로 비워요

CODE

A solid box

```
fill with STONE  
  from (0, 0, 0)  
  to   (4, 3, 4) replace
```

Small corner first, big corner second. The middle number **3** is the **height** (y).

작은 모서리 먼저, 큰 모서리 나중에 — 가운데가 높이

CODE

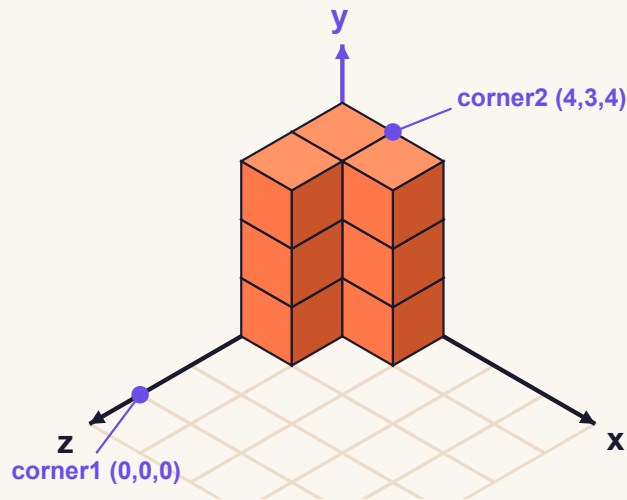
Hollow it out + cut a door

```
fill STONE from (0,0,0) to (4,3,4) # solid box  
fill AIR   from (1,1,1) to (3,2,3) # hollow inside  
fill AIR   from (2,1,0) to (2,2,0) # doorway
```

Line 2 hollows the inside. Line 3 cuts a doorway from the floor up.

3D SPACE

Two corners $\rightarrow$ a hollow room



두 모서리 $(0,0,0)$ 과 $(4,3,4)$ 사이를 채우고, 안을 비우고, 벽에 문을 뚫어요

PREDICT 🤖

What's left after both lines?

Predict first — then click reveal

```
fill stone from (0, 0, 0) to (4, 3, 4)
fill air from (1, 1, 1) to (3, 2, 3)
```

Reveal output 🔍 reveal

VOCAB

New words today

fill	fills everything between two corners
corner	one of the two (x,y,z) ends fill needs
hollow	empty inside, made with fill air
doorway	an opening from the floor up to walk through

채우기

모서리

속이 빈

출입구



DEBUG

What's wrong with this code?

Goal: a **room** you can stand inside.

buggy

```
fill stone from (0, 0, 0) to (6, 4, 6)
```

Think first — then click next.



The bug

Bug: it's a **solid** block of stone — there's no inside. Add a second `fill air` for the inner corners to hollow it out.

fixed

```
fill stone from (0, 0, 0) to (6, 4, 6)
fill air from (1, 1, 1) to (5, 3, 5)
```

YOUR TURN!

Build a hollow room

- ✓ Fill a solid stone box (small corner first)
- ✓ Fill the inside with **air** to make it hollow
- ✓ Cut a doorway starting at the **floor** ($y = 1$)
- ✓ Walk inside to check it works
- ✓ Send a photo of you standing in the room



QUIZ

QUESTION 1 OF 3

What does **fill** need to build a box?

A loop that grows y.

Just an x and a z.

One coordinate and a count.

Two corners, each with x, y, z.



QUIZ

QUESTION 2 OF 3

How do you turn a solid box into a room?

Fill it with more stone.

Fill the inside with air.

Place a gold block on top.

Make it taller.



QUIZ

QUESTION 3 OF 3

In $(2, 5, 3)$, which number is the **height**?

5 (the middle — y)

2 (the first — x)

3 (the last — z)

None of them

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



All three unlocked!

Next: design the whole museum.

다음 주: 미술관 설계도 그리기